

Taxonomies enhance pharmaceutical content intelligence

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Global pharmaceutical organizations are facing increasing competitive and regulatory pressure. It is a business imperative to understand how existing products and those in the pipeline are affected by current research, competitive threats, existing and forthcoming regulations. In today's fast-paced, information-intense environment, this is a highly complex challenge as critical business information resides in multiple repositories, both inside and outside the organization.

In pharmaceutical applications, traditional full-text search engines, entity extraction and summarization tools are insufficient to address the complex scientific concepts associated with the business. In order to support the complex nature of pharmaceutical lexicon, subject area taxonomies can be used to support the complicated information categorization and filtering requirements of medical, regulatory and clinical researchers. The complex concepts contained within a compound's indication can also be used as the basis of categorization for the variety of existing internal and external sources. Unfortunately, the cost of development of taxonomies sufficient to support these queries has been prohibitive in both time and labor.

Building taxonomies is expensive. Most organizations cannot afford large librarian staff, so taxonomy development is typically assigned to an in-house information scientist or external consultants who are subject matter experts (SME) in the business. Regardless of whether an internal or external SME is used, experts estimate that manually creating and training each topic in a taxonomy can require two to three hours and cost as much as US\$300. Additionally, taxonomies require regular maintenance as information and the business dynamics evolve. One solution is pre-

built taxonomies, which may contain thousands of topics and cover a variety of specialized subject areas that are relevant to the business. Commercially available taxonomies can provide taxonomy development teams with an excellent starting point and minimize time and expense of developing them from scratch. These taxonomies may require some editorial modification; however, they can significantly shorten the overall time to completion and expand the breadth of topical coverage.

A good taxonomy should have a hierarchical structure that relates the categories and subcategories of a domain, as well as matching rules or pre-developed training sets. These rules are critical for accurately tagging the concepts found in the unstructured data. Taxonomies can often lead to confusion, intimidation and frustration due to a broad array of commercial offerings, terminology, tools, techniques and approaches. Thesauri, controlled vocabularies, taxonomies and ontologies are often used interchangeably in any 'taxonomy' dialog. While these terms are related, what is delivered is very different. Many publicly available 'taxonomies', such as MeSH, are really controlled vocabularies. While they contain the standard vocabularies for information retrieval within a specific industry and are often delivered in a hierarchical structure, they lack the essential matching rule sets. It is a good rule of thumb to fully understand what is required for a successful implementation before embarking on a taxonomy project.

Intellisophic has developed a series of pharmaceutical and regulatory intelligence taxonomies which, when used within text categorization solutions, can index both internal and external data so that it can be intelligently mined for information relevant to the compa-

ny's products. Information from internal repositories, external databases, regulatory databases, news feeds, journal feeds, or any place information exists that can affect the business can be monitored for change and the appropriate personnel can be alerted to changes in their area.

To develop detailed taxonomies, subject areas must be modeled at a very granular concept level. Deep, rich taxonomies allow for greater accuracy in concept identification. Taxonomic models are available in a number of subject areas such as, medical taxonomies, therapeutic areas, dosage forms, drug manufacturing, drugs and indications, as well as detailed taxonomies in all disease and research areas.

Having a centralized intelligence system can organize all business critical intelligence information related to the product pipeline in one location. A centralized intelligence system yields significant reductions in internal and external research times and increased awareness of critical information related to licensing and regulatory changes.

By implementing a centralized intelligence system containing taxonomies that are frequently updated, a pharmaceutical organization can reduce the risk associated with changing regulatory guidelines; increasing their understanding and rate of awareness. Deploying a system capable of interpreting unstructured data can protect market share and alert professionals to threats from competitive product and licensing initiatives around the globe. Through improved inter-departmental/divisional/subsidiary communication, pharmaceutical organizations can increase efficiency in competitive intelligence gathering, synthesis and analysis. ■